

AASHRAY

Maharashtra Local Emergency Response

End-to-End Pipeline Validation

Final isolated validation run · executed 2026-10-09 (UTC 07:06–07:07) · measured results only

1. Objective

This experiment validates the integrated AASHRAY emergency-response pipeline under realistic, Maharashtra-local conditions. It uses only existing genuine disaster scenarios, real HTTP calls to the AASHRAY service and the unmodified Route Engine, sequential execution, and the existing responder lifecycle. It validates AASHRAY under geographically local Maharashtra emergency-response scenarios.

2. Pipeline and system under test

Incident → Resource requirement → Resource allocation → Demand fulfilment → Mission creation → Agent assignment (AVAILABLE → BUSY) → Route Engine (one call per leg, unique route_engine_mission_id per leg) → Mission lifecycle (complete or cancel; BUSY → AVAILABLE).

System under test: the current AASHRAY code (resource_services) run unmodified on port 8100 against the real data root, and the Route Engine (commit 11cc789, clean working tree) run unmodified on port 3100. Before execution the AASHRAY test suite passed 146 of 146 tests.

3. Dataset selection

- Original genuine-disaster dataset: 100 scenarios (experiments/e2e_100_previous_invalid_distance_scope, read-only; the previous run is not overwritten).
- Maharashtra scenarios: 23. Outside Maharashtra (excluded): 77.
- Maharashtra scenarios excluded for distance > 150 km: 12 of 23. Ten were excluded by the screening lower bound or estimate; two were borderline and excluded by direct measurement (E2E-067 Alibag: 182.1 km; E2E-070 Mahad: 167.7 km). 11 qualify.
- Executed: 11. No scenario was added, merged or re-geocoded. Incident identifiers were renamed (E2E100-INC → MHF-INC) so this run's missions are distinct.

4. Geographic and distance constraints

- Maharashtra locations only; coordinates from the existing dataset, unmodified.
- HARD LIMIT: the measured total route distance of every executed scenario must be ≤ 150 km. Result: 0 scenarios above 150 km.
- Preferred: most scenarios ≤ 50 km. Result: 9 of 11 ≤ 50 km.
- Warehouses (WH001-WH005) and responders (R001-R005) are the canonical files; their locations and statuses were not modified.

5. Execution methodology

- Quota check first: one Google Routes probe returned HTTP 200 at 07:04 UTC. The two borderline final legs were then measured by road (E2E-067: 105.4 km; E2E-070: 163.2 km) and both were excluded.
- Services: Route Engine (3100) and AASHRAY (8100) started; health checks passed; Neon snapshot taken before and after execution.
- Each scenario: POST /orchestrate (real pipeline), invariant checks (runner, 21 invariants), then one lifecycle call: POST /mission/{id}/complete, or /cancel for E2E-023, E2E-059 and E2E-092 (pre-selected).
- Sequential only; no retries; no responder or warehouse reset; no manual database edits. The run would have stopped on any Google quota error (none occurred during execution).
- Four earlier attempts were blocked by the Google per-day quota (2026-10-08 19:51 UTC to 2026-10-09 04:20 UTC). None executed a scenario or called /orchestrate. Each attempt is logged in the README.

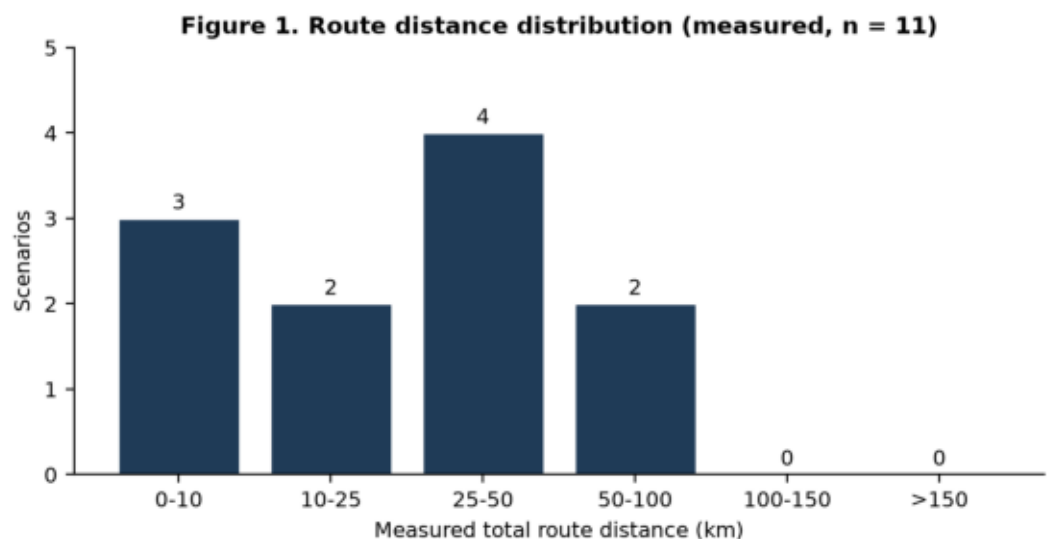


Figure 1. Measured total route distance (n = 11). Buckets: 0-10 km: 3; 10-25: 2; 25-50: 4; 50-100: 2; 100-150: 0; > 150 km: 0.

6. Scenario execution results

ID	Disaster	Final classification	Resp.	WH	Legs	km	min	Lifecycle
E2E-001	1944 Bombay explosion	SUCCESS_PARTIALLY_FULFILLED	R002	5	6	99.58	254.6	complete 200
E2E-023	Maharashtra floods of 2005	SUCCESS_FULLY_FULFILLED	R001	3	4	32.55	136.4	cancel 200
E2E-038	2013 Thane building collapse	SUCCESS_FULLY_FULFILLED	R002	1	2	37.30	86.0	complete 200
E2E-051	2017 Mumbai stampede	SUCCESS_FULLY_FULFILLED	R002	1	2	16.43	71.3	complete 200
E2E-053	Kamala Mills fire	SUCCESS_FULLY_FULFILLED	R002	1	2	17.97	73.3	complete 200
E2E-059	2019 Mumbai foot overbridge c...	SUCCESS_FULLY_FULFILLED	R002	1	2	28.02	62.4	cancel 200
E2E-076	2021 Mumbai landslide	SUCCESS_FULLY_FULFILLED	R002	1	2	9.05	42.1	complete 200
E2E-081	2022 Mumbai building collapse	SUCCESS_FULLY_FULFILLED	R002	1	2	5.31	30.0	complete 200
E2E-087	2023 Raigad landslide	SUCCESS_FULLY_FULFILLED	R002	1	2	75.11	117.1	complete 200
E2E-092	Mumbai billboard collapse	SUCCESS_FULLY_FULFILLED	R002	1	2	9.48	40.9	cancel 200
E2E-093	2024 Thane explosion	SUCCESS_FULLY_FULFILLED	R002	1	2	41.26	95.9	complete 200

7. Resource allocation and fulfilment (measured, 11 scenarios)

Resource	Required	Allocated	Shortage	Fulfilment %
Water	164,010.00	6,760.00	157,250.00	4.1%
Food	82,005.00	3,905.00	78,100.00	4.8%
Medical kits	4,100.25	550.25	3,550.00	13.4%

Every scenario satisfies allocated + shortage = required for each resource (11/11). Fulfilment is low because the stated per-person requirements (water 2, food 1, medical kits 0.05 per affected person) are far larger than the total stock of the five warehouses (water 2,750; food 1,900; medical kits 450 — all of it allocated to E2E-001). Shortage is a legitimate business outcome, not a software failure. E2E-001 is the only partially fulfilled scenario; all other scenarios were fully fulfilled. Warehouse stock is not decremented by the pipeline, and warehouses.json is unchanged after the run, so every scenario was allocated from the same starting stock.

8. Mission creation, responder assignment and lifecycle

Missions created: 11. Responder assignments: 11 (2 responders: R002 × 10, R001 × 1). Reuse of a responder after release: 9 missions. Closed: 8 COMPLETED, 3 CANCELLED; ASSIGNED remaining: 0. Every assigned responder moved AVAILABLE → BUSY on assignment and returned to AVAILABLE on close; the runner confirmed that only the assigned responder was released and that no other responder changed (invariants 14-18, all 11 scenarios).

Lifecycle checks run after execution on the closed missions: 4 of 4 passed (completing a COMPLETED mission again → 200 idempotent; cancelling a COMPLETED mission → 409 MISSION_ALREADY_CLOSED; cancelling a CANCELLED mission again → 200 idempotent; completing a CANCELLED mission → 409 MISSION_ALREADY_CLOSED). responders.json and missions.json were byte-identical across these calls.

9. Route results (measured)

Route Engine calls: 28 (one per leg; 28 successful, 0 failed). Google Routes errors: 0. OpenWeb Ninja requests failed: 28 (HTTP 429, quota). Every leg therefore reports road_condition_status UNKNOWN and no matched incidents. Routing used real road distances and times from Google, but it was NOT disaster-aware. Warehouse sequences were the fixed WH001 → WH005 chain, ordered as the allocation produced them. Per scenario the total equals the sum of its legs (invariant 13). Each leg's route_engine_mission_id is the parent mission ID with -LEG-n (invariant 11). Google requests made inside the Route Engine were not counted separately; the Route Engine returned no Google errors.

Figure 2. Measured total route distance by scenario

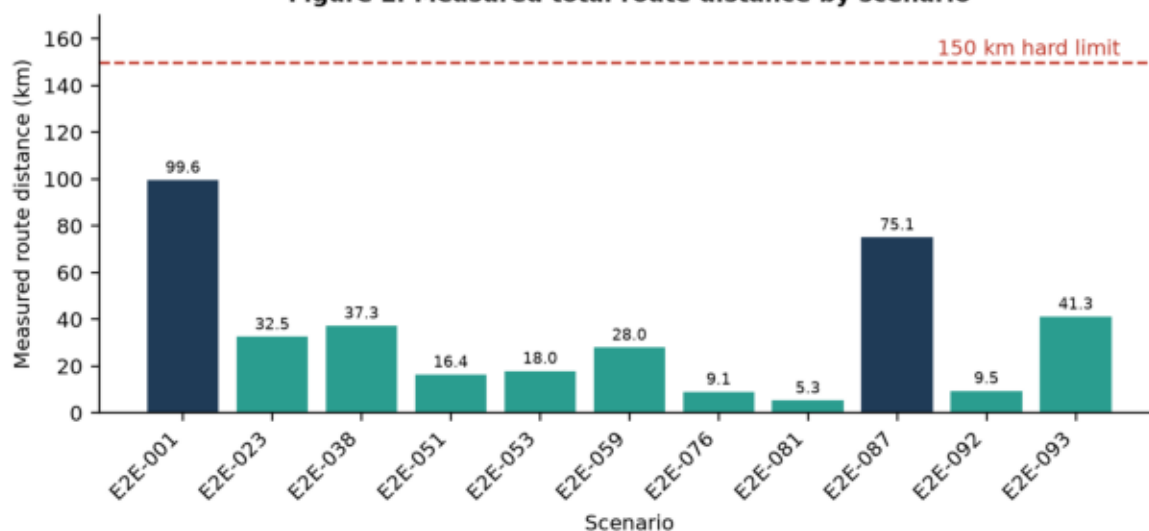


Figure 2. Measured total route distance per scenario against the 150 km hard limit (teal ≤ 50 km, navy 50–100 km).

Figure 3. Resource required vs allocated vs shortage (measured, 11 scenarios)

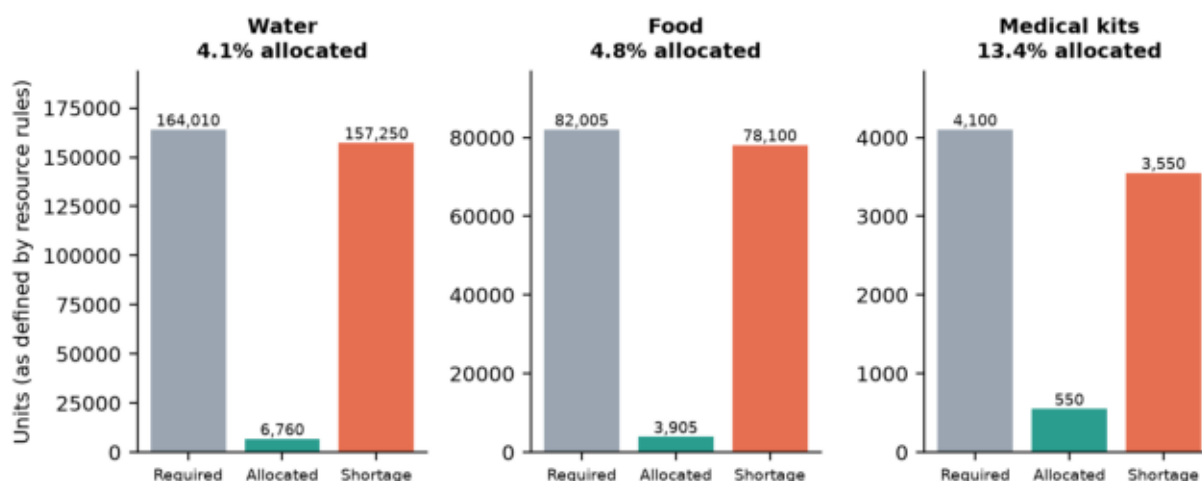


Figure 3. Required, allocated and shortage totals for the 11 scenarios (measured).

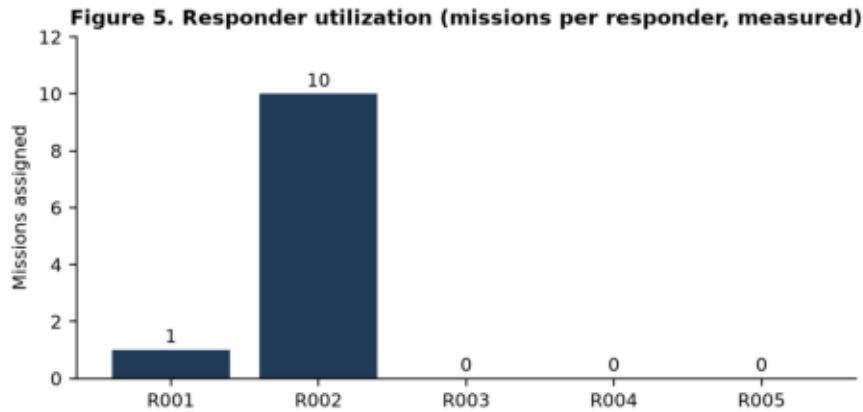
10. Route statistics (measured total route distance and time)

Metric	Distance (km)	Time (min)
Minimum	5.31	30.01
P25	12.96	—
Median	28.02	73.30
Mean	33.82	91.82
P75	39.28	—
Maximum	99.58	254.62
Total	—	1010.02

Distance bucket	0-10	10-25	25-50	50-100	100-150	> 150
Scenarios (measured)	3	2	4	2	0	0

Legs: 28 in total (2.55 per route; 1 warehouse → 2 legs, 3 → 4, 5 → 6). Multi-warehouse chains dominate the distance. E2E-001 (5 warehouses) totals 99.58 km: 79.72 km to WH005 through the warehouse chain, then 19.86 km to the incident. This follows from AASHRAY's fixed, location-blind WH001 → WH005 ordering, not from the incident location; it is reported, not changed.

11. Responder utilization



R001 and R002 were used; R003, R004 and R005 were never assigned. R005 started BUSY (no mission record) and remained BUSY, which is the canonical state. Final state equals initial state: 4 AVAILABLE, 1 BUSY (identical to the initial state for every responder: yes).

12. Failures and legitimate shortages

- Technical failures (PIPELINE_FAILURE / VALIDATION_FAILURE): 0. Route Engine failures: 0. Google quota errors during the run: 0. No NO_SAFE_ROUTE or NO_RESPONDER_AVAILABLE outcomes occurred.
- Legitimate business outcome: 1 scenario SUCCESS_PARTIALLY_FULFILLED (E2E-001, 80,000 affected). It is not counted as a failure; all 11 orchestrations returned COMPLETED.
- External API limitation: OpenWeb Ninja returned HTTP 429 for all 28 requests. This degrades disaster-aware routing and is not an AASHRAY allocation failure.
- Excluded before execution, not failures: E2E-067 (182.1 km measured) and E2E-070 (167.7 km), both above 150 km by measurement. They are reported in §3 and were not executed.

13. Invariant validation

Runner invariants (1-21) on every executed scenario: 0 scenarios with failures. Builder checks on every scenario (11 scenarios): required ≥ allocated; shortage = required – allocated; allocation = sum of warehouse allocations; mission resources = allocated resources; route distance and time = sum of legs; leg count = warehouses + 1; every leg has its parent-derived route_engine_mission_id; scenario is Maharashtra; every route ≤ 150 km; the assigned responder was AVAILABLE before assignment and AVAILABLE after close; no other responder changed; mission closed with the expected state and no ASSIGNED mission left. All passed.

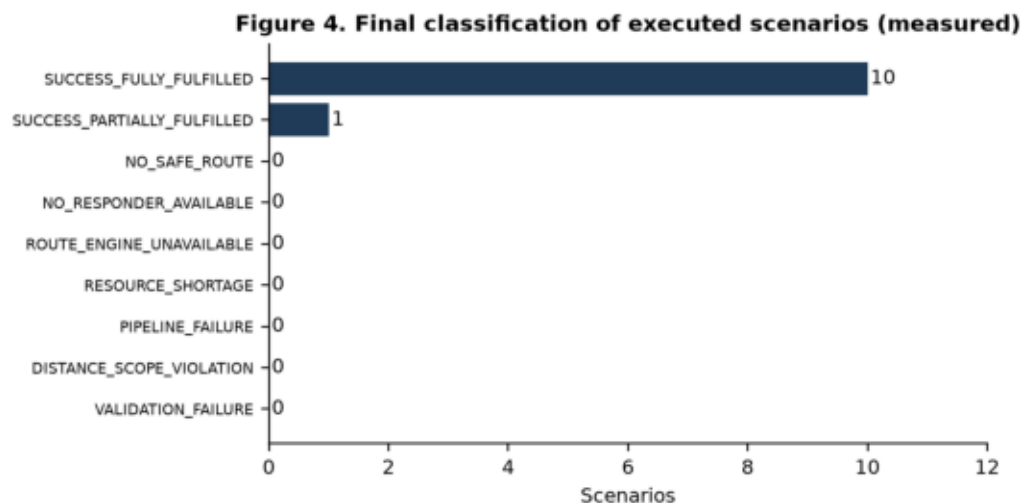


Figure 4. Final classification of the 11 executed scenarios (measured; zero-count categories shown).

14. Database and integrity validation

- Route Engine MissionRouteState rows: 129 before → 157 after (+28 = 28 legs). Every leg has its own row keyed by its route_engine_mission_id.
- RoadConditionEvidence rows: 3 before → 3 after (no new evidence; OpenWeb unavailable).
- warehouses.json unchanged: yes. responders.json byte-identical: yes. missions.json: prior 100 records preserved; 11 new records (8 COMPLETED, 3 CANCELLED); none ASSIGNED.
- 12,000-scenario experiment unchanged: yes. Previous E2E run unchanged: yes. Route Engine repository unchanged: yes. Unexpected AASHRAY file changes: 0.
- Overall post-run integrity check: PASS.

15. Expected (screening) versus measured route distance

ID	Expected basis (screening)	Expected km	Measured km	Diff (km)
E2E-001	measured: full route in the previous run (same chain)	100.5	99.58	-1.0
E2E-023	measured chain + straight-line x 2.0 (estimate)	38.1	32.55	-5.6
E2E-038	measured: full route in the previous run (same chain)	37.4	37.30	-0.1
E2E-051	measured chain + straight-line x 2.0 (estimate)	22.0	16.43	-5.6
E2E-053	measured chain + straight-line x 2.0 (estimate)	23.7	17.97	-5.7
E2E-059	measured chain + straight-line x 1.7 (estimate)	30.9	28.02	-2.9
E2E-076	measured chain + straight-line x 2.0 (estimate)	11.0	9.05	-1.9
E2E-081	measured chain + straight-line x 2.0 (estimate)	6.1	5.31	-0.8
E2E-087	measured chain + straight-line x 1.5 (estimate)	79.5	75.11	-4.4
E2E-092	measured chain + straight-line x 2.0 (estimate)	11.3	9.48	-1.8
E2E-093	measured chain + straight-line x 1.7 (estimate)	50.5	41.26	-9.2

Expected values were screening estimates made before execution. They were used only for screening and are shown here for transparency. Measured values come from this run's Route Engine responses. Where the estimate exceeds the measurement, the estimate was conservative; only the measured figure is reported.

16. Conclusion

- Did the E2E pipeline work? Yes for this dataset: all 11 orchestrations completed through every stage (requirement, allocation, fulfilment, mission, assignment, route, lifecycle), with no technical failures.
- Did resource accounting remain correct? Yes. Allocated + shortage = required for every resource in every scenario, and allocation totals match the warehouse allocations. Shortages were large and are reported as they occurred.
- Did the responder lifecycle work? Yes for these missions. Each assigned responder went AVAILABLE → BUSY, was released on close, and returned to its original state. Lifecycle checks (4/4) passed. Responders were reused 9 times.
- Did realistic local routing work? Measured distances stayed between 5.31 and 99.58 km, with a median of 28.02 km. Routing was real but not disaster-aware, because OpenWeb Ninja was unavailable.
- Were any scenarios over 150 km? Among the executed scenarios, none (0). Two scenarios were measured above

150 km before execution and excluded.

- What failed? Nothing technical. Shortages are business outcomes, and OpenWeb Ninja failed with HTTP 429 on all 28 requests.
- Failures came from: AASHRAY — none. Route Engine — none. External APIs — OpenWeb Ninja quota (HTTP 429). Google Routes — quota blocked the run at 04:20 UTC on 2026-10-09 (before the reset), and the run went ahead after the 07:00 UTC reset.

17. Limitations

- Scale: 11 scenarios from one region, all in Maharashtra. These results do not show that AASHRAY is universally validated.
- Disaster-aware routing was not tested: OpenWeb Ninja returned HTTP 429 on all requests, so road conditions are UNKNOWN.
- Stock was not depleted between scenarios; each scenario drew from the same starting stock. Sequential depletion, competing demand and concurrency were not tested.
- Google routes are traffic-aware. Measured distances and times vary between runs. E2E-001 measured 99.58 km here against 100.54 km in the previous run.
- Only one responder assignment per mission; multi-responder dispatch was not tested.
- The two borderline scenarios were excluded by measurement, not executed.

18. What was validated and what was not

Validated by this run: the real AASHRAY HTTP pipeline on 11 Maharashtra scenarios; resource accounting (allocated + shortage = required); mission creation and warehouse pickup order; responder assignment and lifecycle release, reuse and idempotency; route legs with unique Route Engine IDs and totals; database recording of legs; integrity of data files and the 12,000-scenario experiment.

Not validated: disaster-aware routing (OpenWeb unavailable); depletion of stock across sequential missions; concurrent scenarios; re-routing after road-condition changes; scenarios beyond the 11 selected; behaviour under Google quota exhaustion mid-run (the run stopped at the probe; it did not occur mid-run).

Overall result: PASS WITH LIMITATIONS. Reason: every executed scenario completed with its invariants satisfied, but disaster-aware routing could not be validated and the dataset is small.